Yijin Zeng

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Summary

Final year PhD candidate in Statistics and Machine Learning at Imperial College London.

– Research focuses on developing new methodologies for statistical hypothesis testing, statistical correlation, and change point detection in the presence of missing data.

– Applications include A/B testing, causal inference, information retrieval, and continual learning.

– Strong statistics and machine learning theoretical foundations with hands on experience in end-to-end data science, machine learning, and deep learning projects.

– Proficient in Python, R and SQL. Solid understanding of data structures and algorithms.

– Objective: currently looking for a data scientist/machine learning researcher role in industry.

Education

- **Imperial College London**, London, UK — Oct. 2021– Oct. 2025 (Estimated)
  PhD in Statistics and Machine Learning (StatML CDT, Imperial & University of Oxford)
  – *Funded by Roth Scholarship*
- **Imperial College London**, London, UK — Oct. 2020 – Sep. 2021
  MSc in Statistics
  – *Degree classification: Distinction*
- **Beijing Institute of Technology**, Beijing, China — Sep. 2016 – Jun. 2020
  BSc in Mathematics
  – *Top 10% of cohort; Thesis with Distinction*

Work Experience

TESCO Data Scientist PhD Internship (London) — Aug.2024 – Oct.2024

– Worked with forecasting team and supply chain team, delivering an end-to-end machine learning pipeline for demand forecasting, including feature selections, feature engineering, data cleaning, pre-processing, model training, and evaluation.

– Designed and trained multiple deep neural network architectures for supply chain forecasting, achieving accuracy comparable or higher (5% $\sim$ 10% depending on the product categories) to TESCO's current approach with less workflows.

IQVIA Data Analyst Internship (Beijing) — Jul.2019 – Sep.2019

– Cleaned and analyzed survey and experimental datasets, performing feature engineering, missing-data handling, and statistical testing for supporting final conclusions.

– Worked on data visualization for efficiently communicating numerical results to managers and non-technical stakeholders.

Publications

- **Zeng Y**, Adams NM, Bodenham DA. On two-sample testing for data with arbitrarily missing values. arXiv preprint arXiv:2403.15327. 2024 Mar 22.
  – *A two-sample hypothesis testing method for partially observed univariate data to detect location shifts, with no assumption of the values of missing data.*

- **Zeng Y**, Adams NM, Bodenham DA. MMD Two-sample Testing in the Presence of Arbitrarily Missing Data. arXiv preprint arXiv:2405.15531. 2024 May 24.
  - *Extended our framework for high-dimensional data with any distributional shifts.*
- **Zeng Y**, Adams NM, Bodenham DA. Exact Bounds of Spearman's footrule in the Presence of Missing Data with Applications to Independence Testing. arXiv preprint arXiv:2501.11696. 2025 Jan 20.
  - *Tight bounds for Spearman's footrule in the presence of missing data, with applications for independence testing.*

Open Source Packages

- **Zeng Y**, Bodenham D, and Adams N. Package 'wmwm'.
  - *R and Python packages for performing Wilcoxon-Mann-Whitney two sample hypothesis testing in the presence of missing data.*
- **Zeng Y**. Package 'bosfr'.
  - *R package for computing tight bounds of Spearman's footrule in the presence of missing data.*

Selected Data Science Projects

- E-Commerce Order Cancellation Prediction
  - *Predict order cancellation with highly imbalanced dataset for real-world impact.*
- Deep Learning Hangman Game
  - *Play word guessing games using deep learning with high success rate (∼60%).*
- Stock Clustering
  - *Cluster stocks using daily return to produce balanced investment groups.*

Selected Advanced Courses

- Machine Learning in Finance (by Dr. Mikko Pakkanen)
  - *Machine learning algorithms for high-frequency trading (HFT)*
- Computational Optimal Transport and Statistical Learning with Missing Values (by Prof. Gabriel Peyré & Dr. Julie Josse)
  - *Optimal transport for machine learning; dealing with missing data in machine learning*
- Selective Inference (by Dr. Daniel Garcia Rasines & Prof. Alastair Young)
  - *Valid inference for the 'post selection' problem in modern statistics and machine learning*

Selected Teaching Experience

- Probability for Statistics (MSc) — Summer term 2023
- Data Science (MSc) — Spring term 2024
- Statistical Clinics (MSc) — Summer term 2024
- Probability and Statistics (BSc) — Spring term 2024

Skills

- Language: Chinese (Native), English (Proficient).
- Programming & Tools: Python, R, SQL, TensorFlow, pandas, NumPy, scikit-learn, MATLAB; Azure ML; HPC; Linux; Git.